Balance collapse, screening, and the d^2/ACAT switch

A simulation memo for the RItools team

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These simulations settle two practical questions about `balanceTest()`: (1) what does the d^2 omnibus actually respond to, so we know what to tell users it means; and (2) are the two proposed changes – a screening step for near-exactly-matched covariates, and an automatic switch to the Cauchy (ACAT) omnibus – worth building, and for what reason.

The reusable functions are in `vignettes/sim-helpers.R` (sourced above); every experiment's own code is shown in the chunk that produces its result.

The data-generating process and the knobs

Every experiment uses matched **pairs** (one treated, one control per set), because pairs make the algebra transparent and because the matched-pair case is where the collapse is sharpest. There are k covariates, $x_1, \dots, x_k$, and they are **correlated** across columns. For each pair s and covariate j:

- a pair **location** m_{sj} – the BETWEEN-pair variation, the part of covariate j that stratification removes. It plays no role in any test below, and it is drawn independently of the within-pair difference.
- a treated-minus-control **difference** $D_{sj} = \text{noise}_{sj} + \mu_j$, where noise_{sj} is symmetric (random sign, pair to pair) with marginal SD $\sigma_c[j]$, and μ_j is a constant added to every pair on covariate j.
- the treated unit's value is $m_{sj} + D_{sj}/2$, the control's is $m_{sj} - D_{sj}/2$.

`mu` and `sigma_c` are length-k **vectors**: `mu[j]` is the systematic tilt on covariate j, `sigma_c[j]` is its within-pair noise SD. The covariates are not exactly standardized: each location has unit variance and each within-pair noise has marginal SD `sigma_c[j]` (= 1 everywhere except near-exactly-matched covariates, where it is ~0), so each covariate sits on a roughly unit scale and `mu[j]` is approximately in SD units. The quantity the test reads is $\mu_j/\sigma_c[j]$ (the "standardized bias"). Correlation is moderate equicorrelation (every pair of covariates 0.5), applied to both the locations and the within-pair differences, with column scaling that preserves each marginal SD; `rho` is one tunable argument (`rho = 0` recovers independent covariates).

The data generator:

```
make_pairs
#> function(S, sigma_c, mu, tscale = 1, rho = 0.5) {
#>   k <- length(sigma_c)
#>   stopifnot(length(mu) == k)
#>   ## Covariates are CORRELATED across columns (not independent): moderate
#>   ## equicorrelation R[j,l] = rho off the diagonal, 1 on it (rho = 0 would
#>   ## recover independent covariates).  We apply the same correlation to the
#>   ## between-pair locations and to the within-pair differences; it is the
#>   ## correlation of the within-pair DIFFERENCES that puts off-diagonal entries
#>   ## into V_d and so actually reaches the tests.  Column scaling by sigma_c keeps
#>   ## each covariate's marginal within-pair SD equal to sigma_c[j], so the
#>   ## mu/sigma_c ("standardized bias") interpretation is unchanged.
#>   R <- matrix(rho, k, k); diag(R) <- 1
#>   L <- t(chol(R))                                          # L %*% z has cov R
#>   cor_rows <- function() t(L %*% matrix(rnorm(k * S), nrow = k))  # S x k, row cov R
#>   m     <- cor_rows()                                      # pair locations
#>   noise <- cor_rows() *
#>            matrix(sigma_c * tscale, nrow = S, ncol = k, byrow = TRUE)
#>   D     <- noise + matrix(mu * tscale, nrow = S, ncol = k, byrow = TRUE)  # tr - ctl
#>   treated <- m + D / 2
#>   control <- m - D / 2
#>   X <- matrix(0, nrow = 2 * S, ncol = k)
#>   X[seq(1, 2 * S, by = 2), ] <- treated                    # odd rows treated
```

```
#>   X[seq(2, 2 * S, by = 2), ] <- control                        # even rows control
#>   out <- data.frame(z = rep(c(1L, 0L), times = S),
#>                     pair = factor(rep(seq_len(S), each = 2)))
#>   for (j in seq_len(k)) out[[paste0("x", j)]] <- X[, j]
#>   out
#> }
```

Two short helpers used inline below: `scale_within()` applies the size knob t (scale the whole within-pair difference by t, direction fixed), and `smd()` is the Love-plot standardized mean difference (mean difference over pooled SD).

```
scale_within <- function(df, t) {
  for (cn in covariate_names(df)) {
    xt <- df[[cn]]; pm <- ave(xt, df$pair)
    df[[cn]] <- pm + (xt - pm) * t          # scale within-pair part, keep pair mean
  }
  df
}
smd <- function(df, cn) {
  x <- df[[cn]]; g1 <- df$z == 1; g0 <- df$z == 0
  (mean(x[g1]) - mean(x[g0])) / sqrt((var(x[g1]) + var(x[g0])) / 2)   # pooled-SD SMD
}
```

What the test responds to

The d^2 omnibus is $d^2 = \bar{d}' V_d^{-1} \bar{d}$, with $\bar{d}$ the average treated-minus-control difference and V_d its covariance under within-pair re-randomization. With this DGP, $\bar{d} \approx \mu$ (the noise averages out) and $V_d \approx (1/S)$ (within-pair covariance), so for a single covariate

$$d^2 \;\approx\; S\left(\frac{\mu}{\sigma_c}\right)^2.$$

The omnibus responds to the systematic tilt **measured in units of the noise** (μ/σ_c) times the sample size S. Under the null it is central chi-square on rank(V_d) df; under a tilt, noncentral chi-square with $\lambda \approx S\,(\mu/\sigma_c)^2$. With correlated covariates the multivariate version is $\lambda = \mu' V_d^{-1} \mu$, and the alignment of μ with V_d's eigenvectors matters – which is what makes B4 flip.

Three consequences (= A1, A2, A3): scale μ and σ_c together and the ratio is unchanged, so d^2 does not move (A1, the collapse); shrink μ with σ_c fixed and d^2 falls (A2); hold the ratio and grow S and power rises (A3).

Two standardizations – why this differs from a Love plot

The Love-plot standardized mean difference divides by the **pooled SD**, which is dominated by between-pair variation that the size knob t leaves alone – so the SMD scales with t and registers when groups get more or less different. d^2 divides by the **within-pair null SD** (the σ_c that t

shrinks), so the t cancels. The Love plot sees the size of imbalance; d^2 does not. They agree only when σ_c is held fixed (which is what A2 does).

Study 1: what the omnibus does and does not respond to

A1 – the collapse: d^2 ignores the size of the imbalance

One fixed data set with a tilt on x_1; increase or decrease the size of the whole within-pair imbalance by t (direction fixed) over five orders of magnitude.

```r
set.seed(1)
base_df <- make_pairs(S = 100, sigma_c = rep(1, 4), mu = c(0.5, 0, 0, 0))
a1 <- do.call(rbind, lapply(c(100, 10, 1, 0.1, 0.01, 0.001), function(t) {
  r <- run_omnibus(scale_within(base_df, t))
  data.frame(t = t, chisq = round(r$chisq, 3), df = r$df,
             p_d2 = signif(r$p_d2, 3), p_acat = signif(r$p_acat, 3))
}))
a1
```

t	chisq	df	p_d2	p_acat
1e+02	29.307	4	6.8e-06	0.000234
1e+01	29.307	4	6.8e-06	0.000234
1e+00	29.307	4	6.8e-06	0.000234
1e-01	29.307	4	6.8e-06	0.000234
1e-02	29.307	4	6.8e-06	0.000234
1e-03	29.307	4	6.8e-06	0.000234

A thousand-fold change in how different the treated and control units actually are produces no change in the statistic or its p-value. This is the collapse, exact: d^2 is homogeneous of degree zero in the imbalance (numerator scales by t, V_d by t^2, $t \cdot t/t^2 = 1$). A Love plot would show balance improving as t falls; d^2 does not.

Two distinct limits show up, and only one is escapable. **ACAT is invariant too** (`p_acat` flat): it combines per-covariate p-values, each itself a size-blind within-pair test, so the combination is size-blind. Nothing on the within-pair reference sees the size of imbalance. The **other** limit, which ACAT does escape, is the cap on how many covariates d^2 can test: d^2 inverts V_d, whose rank cannot exceed the within-block residual df; past that d^2 freezes (p near 0.45). ACAT never inverts anything (shown in B4).

A2 – the test IS informative about systematic confounding

Hold the within-pair noise fixed (`sigma_c = 1`) and shrink only the systematic tilt `mu` on one covariate from 0.5 to 0.

```r
set.seed(20260617)
mu_grid <- c(0.5, 0.3, 0.2, 0.1, 0.05, 0)
a2 <- do.call(rbind, lapply(mu_grid, function(mu1) {
  ps <- replicate(NREP, run_omnibus(make_pairs(100, rep(1, 3), c(mu1, 0, 0)))$p_d2)
  data.frame(mu = mu1, rejection_rate = rejection_rate(ps),
             median_p = round(median(ps, na.rm = TRUE), 4))
}))
a2
```

mu	rejection_rate	median_p
0.50	1.000	0.0000
0.30	0.856	0.0034
0.20	0.499	0.0503
0.10	0.122	0.2988
0.05	0.072	0.4149
0.00	0.051	0.5041

Unlike A1, the test moves, and correctly. A large tilt is detected almost always; as the tilt vanishes the rejection rate falls to the nominal 0.05 and the p-values become uniform (median ~0.5). So the omnibus is a sound test of the systematic, direction-carrying part of imbalance. A1 and A2 together: it reads μ/σ_c. A1 moved both (ratio fixed -> no response); A2 moved μ alone (ratio falls -> response). "Collapse" is specific to the size dimension.

A3 – more data detects a fixed bias more often

Hold the standardized bias fixed at $\mu/\sigma_c = 0.15$ and grow the study from 25 to 400 pairs (ESS = pairs).

```r
set.seed(20260617)
a3 <- do.call(rbind, lapply(c(25, 50, 100, 200, 400), function(S) {
  ps <- replicate(NREP, run_omnibus(make_pairs(S, rep(1, 3), c(0.15, 0, 0)))$p_d2)
  data.frame(ESS = S, rejection_rate = rejection_rate(ps))
}))
a3
```

ESS	rejection_rate
25	0.072
50	0.141
100	0.273
200	0.564
400	0.894

With the same real bias throughout, power climbs from ~0.10 to ~0.90: ordinary power, $\lambda \approx S\,(\mu/\sigma_c)^2$. The flip side is the warning: because power against any fixed nonzero bias keeps rising with ESS, a large enough matched sample rejects on a bias too small to matter. That "balance tests reject in large samples" problem is a second reason – on top of the collapse – to report a fixed-scale magnitude rather than read the design off the omnibus p.

A4 – why a p-value-vs-ESS plot is jagged

One fixed population of 200 confounded pairs (a modest tilt on x_1). Prune pairs worst-offenders-first (most aligned with the bias), recompute the omnibus at each step, and track the p-value and the statistic against ESS.

```
set.seed(7)
S0 <- 200
df_full <- make_pairs(S = S0, sigma_c = rep(1, 4), mu = c(0.10, 0, 0, 0))
align <- tapply(df_full$x1 * ifelse(df_full$z == 1, 1, -1), df_full$pair, sum)
prune_order <- order(align, decreasing = TRUE)          # drop most-aligned first
lvl <- levels(df_full$pair)
path <- do.call(rbind, lapply(seq(S0, 30, by = -5), function(nkeep) {
  keep <- prune_order[(S0 - nkeep + 1):S0]              # keep the least-aligned nkeep
  sub  <- droplevels(df_full[df_full$pair %in% lvl[keep], ])
  r <- run_omnibus(sub)
  std_bias <- abs(mean(tapply(sub$x1 * ifelse(sub$z == 1, 1, -1), sub$pair, sum)))
  data.frame(ess = effective_sample_size(sub), df = r$df, chisq = r$chisq,
             p_d2 = r$p_d2, p_acat = r$p_acat, std_bias = std_bias)
}))
## representative rows
path[path$ess %in% c(200, 195, 185, 180, 100, 30),
     c("ess", "chisq", "p_d2", "p_acat", "std_bias")]
```

	ess	chisq	p_d2	p_acat	std_bias
1	200	7.586524	0.1079539	0.7828210	0.0457268
2	195	6.595364	0.1588800	0.1284773	0.0274234
4	185	8.353032	0.0794694	0.0232339	0.1299252
5	180	13.803947	0.0079478	0.0028298	0.1727364
21	100	63.999066	0.0000000	0.0000000	0.7872708
35	30	28.384942	0.0000104	0.0000004	1.5831793

As ESS falls the p-value is non-monotone: it ticks up near the full sample (where pruning removes the targeted tilt) and then crashes. The honest driver is the **statistic itself** – d^2 is a hump-shaped function of ESS (low at the full sample, peaking at intermediate ESS as pruning overshoots and induces imbalance, then falling as the sample shrinks). Since $p = \text{pchisq}(d^2, \text{df}, \text{lower} = F)$, the p-value is the monotone transform of that hump.

```
op <- par(mfrow = c(2, 2), mar = c(4, 4, 3, 1))
plot(a2$mu, a2$rejection_rate, type = "b", pch = 19, ylim = c(0, 1),
     xlab = "systematic bias mu (sigma_c = 1)", ylab = "rejection rate (alpha=.05)",
```

```
     main = "A2: p tracks systematic bias")
abline(h = 0.05, lty = 2, col = "grey50")
plot(a3$ESS, a3$rejection_rate, type = "b", pch = 19, ylim = c(0, 1),
     xlab = "effective sample size (= pairs)", ylab = "rejection rate (alpha=.05)",
     main = "A3: ESS buys power")
abline(h = 0.05, lty = 2, col = "grey50")
plot(path$ess, path$p_d2, type = "b", pch = 19, ylim = c(0, 1),
     xlab = "effective sample size", ylab = "omnibus p-value",
     main = "A4: p vs ESS (non-monotone)")
lines(path$ess, path$p_acat, type = "b", pch = 1, col = "red")
legend("topleft", c("d^2", "ACAT"), pch = c(19, 1), col = c("black", "red"), bty = "n")
plot(path$ess, path$chisq, type = "b", pch = 19, col = "blue",
     xlab = "effective sample size", ylab = "omnibus d^2 statistic (df=4)",
     main = "A4 driver: the d^2 statistic vs ESS")
```

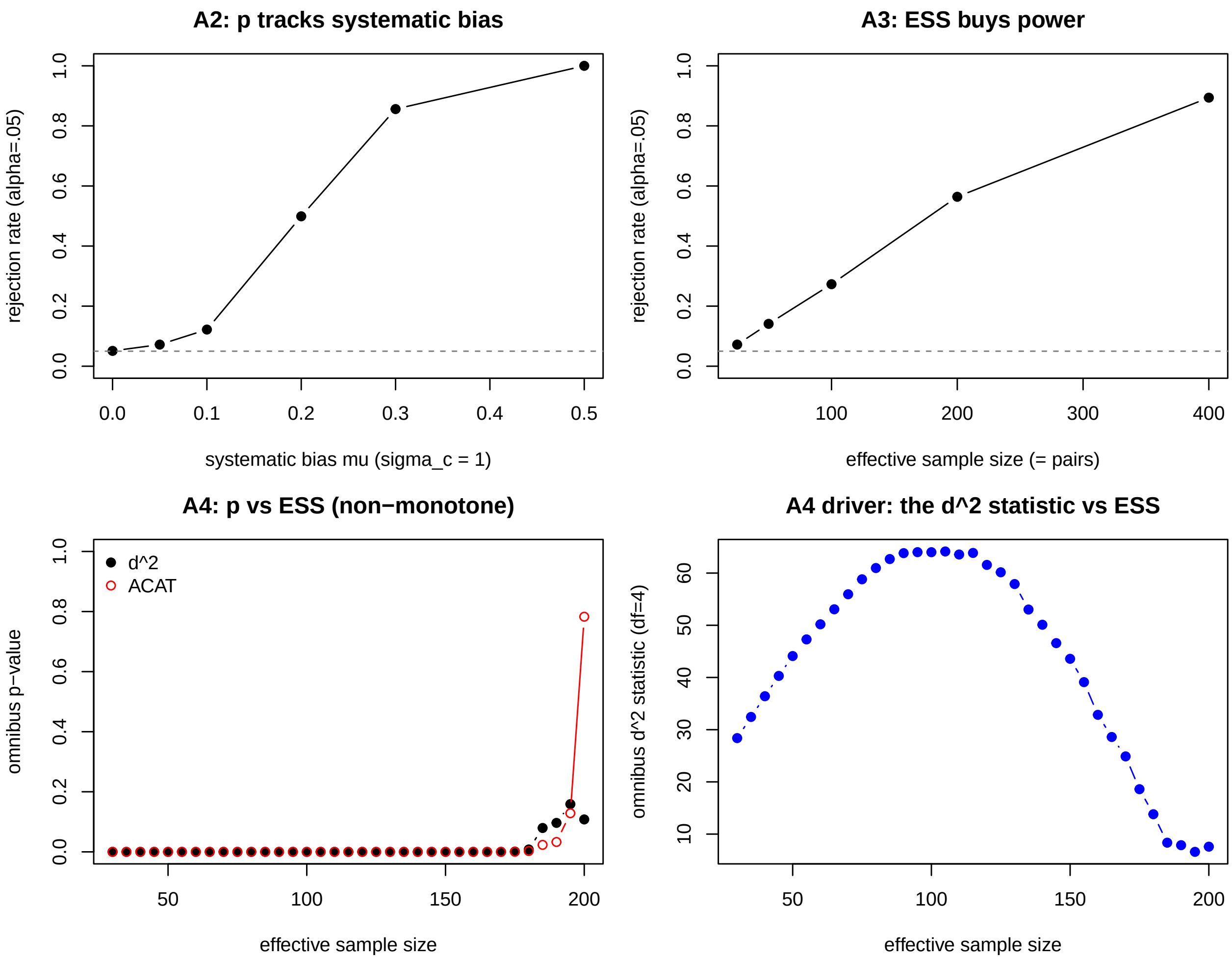

Figure 1: Study 1. A2: p tracks systematic bias. A3: power grows with ESS. A4: p vs ESS is non-monotone (the p-axis crushes it near 0); the de-compressed d^2 statistic shows the hump that drives it.

```r
par(op)
```

The payoff: $d^2 \approx S\,(\text{standardized imbalance})^2$, and along a pruning path both factors move at once. You cannot read balance quality off a p-vs-ESS curve – its shape reflects which pairs you dropped as much as how balanced the survivors are.

Study 2: screening, and the d^2/ACAT switch

There are **two different kinds of screen**, and keeping them distinct is the main lesson:

- A **design screen** asks "is this covariate (or direction) exactly matched?" The prototype is w_j = (within-stratum var)/(total var) per covariate, which exact matching drives to ~0. It depends on covariate values and strata only – not the treatment vector – so it is ungameable (B5). The direction version (B6) generalizes it to linear combinations. It fixes the degenerate-direction pathology but is blind to the size of imbalance (B3).
- A **magnitude screen** asks "is the observed imbalance substantively large?" The prototype is the Love-plot rule: flag covariates with $|\text{SMD}| \geq 0.25$ (see "Provenance of the 0.25-SD rule" at the end of this section for the citation, which is *not* Rosenbaum-Rubin 1985 as often claimed). It uses the pooled-SD scale, so it does NOT collapse (B7); it is the constructive form of "lead with magnitude."

B1 – near-degenerate covariates dilute d^2; the design screen sharpens it

Five covariates: x_1, x_2 near-exactly matched (`sigma_c = 1e-4`, no tilt), a real confound on x_3, x_4, x_5 vary with no tilt. Detect x_3 three ways.

```r
set.seed(20260618)
b1 <- replicate(NREP, {
  df <- make_pairs(100, c(1e-4, 1e-4, 1, 1, 1), c(0, 0, 0.30, 0, 0))
  r_all <- run_omnibus(df)
  r_scr <- run_omnibus(df, covs = screen_survivors(df, tau = 0.01))
  c(p_all = r_all$p_d2, df_all = r_all$df,
    p_scr = r_scr$p_d2, df_scr = r_scr$df, p_acat = r_all$p_acat)
})
b1 <- as.data.frame(t(b1))
data.frame(
  method  = c("d^2 (all 5)", "d^2 (design-screened)", "ACAT (all 5)"),
  power   = round(c(rejection_rate(b1$p_all), rejection_rate(b1$p_scr),
                   rejection_rate(b1$p_acat)), 3),
  mean_df = round(c(mean(b1$df_all), mean(b1$df_scr), NA), 2))
```

method	power	mean_df
d^2 (all 5)	0.837	5
d^2 (design-screened)	0.857	3

method	power	mean_df
ACAT (all 5)	0.659	NA

The two dead covariates consume df and dilute the chi-square; screening them out raises d^2 power. ACAT trails here because it spreads weight over five marginal p-values, two of them pure noise: when the signal is concentrated and the covariance is informative, the chi-square is the better omnibus.

B1b – graceful degradation when everything is (near-)exactly matched

```
set.seed(11)
df_ax <- make_pairs(100, rep(1e-9, 4), rep(0, 4))
r_all <- run_omnibus(df_ax)
surv  <- screen_survivors(df_ax, tau = 0.01)
cat(sprintf("d^2 (all): errored=%s, p=%s   <- hard stop() at R/utils.R:381\n",
            r_all$errored, format(r_all$p_d2)))
#> d^2 (all): errored=TRUE, p=NA   <- hard stop() at R/utils.R:381
cat(sprintf("design screen: %d survivors -> %s\n", length(surv),
            if (length(surv) == 0) "ABSTAIN, no error" else "test survivors"))
#> design screen: 0 survivors -> ABSTAIN, no error
cat(sprintf("ACAT: p=%s   <- NA, no error\n", format(r_all$p_acat)))
#> ACAT: p=NA   <- NA, no error
```

The current d^2 path throws when every covariate is degenerate. The screen turns this into an honest non-result; ACAT returns NA cleanly. The package fix is graceful degradation instead of erroring.

B2 – screening does not distort the type-I error rate

```
set.seed(20260618)
b2 <- replicate(NREP, {
  df <- make_pairs(100, c(1e-4, 1e-4, 1, 1, 1), rep(0, 5))
  r_all <- run_omnibus(df)
  r_scr <- run_omnibus(df, covs = screen_survivors(df, tau = 0.01))
  c(p_all = r_all$p_d2, p_scr = r_scr$p_d2, p_acat = r_all$p_acat)
})
b2 <- as.data.frame(t(b2))
data.frame(method = c("d^2 (all)", "d^2 (screened)", "ACAT"),
           size_nominal_0.05 = round(c(rejection_rate(b2$p_all),
                                       rejection_rate(b2$p_scr),
                                       rejection_rate(b2$p_acat)), 3))
```

method	size_nominal_0.05
d^2 (all)	0.054
d^2 (screened)	0.050
ACAT	0.050

All three hold their size: the screen uses a design quantity, not the observed imbalance, so it does not inflate false positives.

B3 and B7 – design vs magnitude screen, read against A1 on the SAME data

A1, B3, and B7 use the **identical** data set (the A1 base, `seed 1`) and the **identical** t-sweep, so they read row for row. A1 is the omnibus with no screen; B3 adds the design screen (w_j, `tau = 0.01`); B7 adds the magnitude screen (flag $|\text{SMD}| \geq 0.25$).

```
set.seed(1)
ab_base <- make_pairs(100, rep(1, 4), c(0.5, 0, 0, 0))
trio <- do.call(rbind, lapply(c(100, 10, 1, 0.5, 0.25, 0.1, 0.01), function(t) {
  df_t <- scale_within(ab_base, t)
  surv <- screen_survivors(df_t, tau = 0.01)          # design screen
  r    <- run_omnibus(df_t, covs = surv)
  smds <- vapply(covariate_names(df_t), function(cn) smd(df_t, cn), numeric(1))
  data.frame(
    t            = t,
    A1_omnibus_p = signif(run_omnibus(df_t)$p_d2, 3),
    B3_survivors = sprintf("%d/4", length(surv)),
    B3_screen_p  = if (is.na(r$p_d2)) "ABSTAIN" else signif(r$p_d2, 3),
    B7_SMD_x1    = round(smds[["x1"]], 3),
    B7_flagged   = sprintf("%d/4", sum(abs(smds) >= 0.25)))
}))
trio
```

t	A1_omnibus_p	B3_survivors	B3_screen_p	B7_SMD_x1	B7_flagged
100.00	6.8e-06	4/4	6.77e-06	0.873	2/4
10.00	6.8e-06	4/4	6.77e-06	0.859	2/4
1.00	6.8e-06	4/4	6.77e-06	0.417	1/4
0.50	6.8e-06	4/4	6.77e-06	0.229	0/4
0.25	6.8e-06	4/4	6.77e-06	0.118	0/4
0.10	6.8e-06	0/4	ABSTAIN	0.047	0/4
0.01	6.8e-06	0/4	ABSTAIN	0.005	0/4

B3 (design screen is NOT a way around the collapse). Where the screen keeps covariates (large t), B3's p equals A1's collapsed value exactly. As t shrinks the within-stratum variance fraction falls below `tau` and the screen ABSTAINS – it never returns a p that rewards the smaller imbalance.

B7 (the Love-plot screen IS the non-collapsing one). `SMD(x1)` tracks the size of imbalance, crossing 0.25 between $t = 1$ and $t = 0.5$. At a well-matched design ($t = 0.1$) the three verdicts diverge sharply: A1 still rejects (the collapse), B3 abstains, and B7 reports `SMD = 0.05 < 0.25` = balanced – the substantively correct answer. It escapes the collapse because the SMD uses the fixed pooled-SD scale. Using the Love plot to screen is the constructive form of "lead with magnitude"; the 0.25 threshold is an equivalence margin.

Two honest costs of the magnitude screen: (i) it is assignment-dependent (reads the observed imbalance), so not ungameable like w_j; and (ii) you cannot screen by observed SMD and then trust the within-pair omnibus p on the survivors – selecting imbalanced covariates inflates that test. Use it as a magnitude/ equivalence **report**, not a pre-filter feeding the test.

B6 – the design screen should work on DIRECTIONS, not covariates

The per-covariate w_j misses a degenerate **linear combination**. Here x_1, x_2 each vary within pairs (so each w_j is moderate) but share the same within-pair difference, so the $x_1 - x_2$ direction is near-exactly matched; a real confound sits on x_3. The direction screen (`sv_screen_project` in `sim-helpers.R`) generalizes w_j to the generalized eigenvalues of within-vs-total covariance.

```
make_degen_combo <- function(S, eps = 5e-3) {
  df <- make_pairs(S, rep(1, 4), c(0, 0, 0.30, 0), rho = 0.3)
  dev1 <- df$x1 - ave(df$x1, df$pair)
  df$x2 <- ave(df$x2, df$pair) + dev1 + rnorm(nrow(df), sd = eps)  # x2 shares x1's within-pai
  df
}
set.seed(61)
d6 <- make_degen_combo(120)
round(within_var_fraction(d6), 4)                  # per-covariate w_j: all moderate
#>     x1     x2     x3     x4
#> 0.4376 0.4220 0.4942 0.3900
round(sv_screen_project(d6, tau = 0.01)$lambda, 5) # direction eigenvalues: one ~0
#> [1] 0.63965 0.48969 0.38090 0.00002
```

The per-covariate w_j are all moderate (so that screen drops nothing) while the direction screen finds one eigenvalue ~0 (the $x_1 - x_2$ combination) and drops it. Power to detect x_3:

```
set.seed(62)
b6 <- replicate(NREP, {
  df <- make_degen_combo(120)
  c(p_all = run_omnibus(df)$p_d2,
    p_pc  = run_omnibus(df, covs = screen_survivors(df, tau = 0.01))$p_d2,
    p_sv  = run_sv_screen(df, tau = 0.01)$p_d2,
    p_acat = run_omnibus(df)$p_acat)
})
b6 <- as.data.frame(t(b6))
data.frame(method = c("unscreened d^2", "per-covariate screen",
                      "direction screen", "ACAT"),
```

```
         power  = round(c(rejection_rate(b6$p_all), rejection_rate(b6$p_pc),
                          rejection_rate(b6$p_sv), rejection_rate(b6$p_acat)), 3))
```

method	power
unscreened d^2	0.802
per-covariate screen	0.802
direction screen	0.848
ACAT	0.787

The per-covariate screen misses the degenerate combination (drops nothing, same power as unscreened); the direction screen catches it and gives a sharper test. The production screen should work on the singular values of V_d.

B4 – keep both omnibuses: under correlation ACAT is the more robust one

Three alternatives. "dense" = small tilt on all 8 covariates; "sparse" = one strong tilt among 8; "highdim" = one strong tilt among 20 covariates with only 15 pairs (so #covariates exceeds rank(V_d)).

```
set.seed(20260618)
scen <- list(
  dense   = list(S = 100, sigma_c = rep(1, 8),  mu = rep(0.12, 8)),
  sparse  = list(S = 100, sigma_c = rep(1, 8),  mu = c(0.55, rep(0, 7))),
  highdim = list(S = 15,  sigma_c = rep(1, 20), mu = c(0.9, rep(0, 19))))
b4 <- do.call(rbind, lapply(names(scen), function(nm) {
  s <- scen[[nm]]
  out <- replicate(NREP, {
    r <- run_omnibus(make_pairs(s$S, s$sigma_c, s$mu)); c(r$p_d2, r$p_acat) })
  data.frame(scenario = nm, k = length(s$sigma_c), S = s$S,
             power_d2 = round(rejection_rate(out[1, ]), 3),
             power_acat = round(rejection_rate(out[2, ]), 3))
}))
b4
```

scenario	k	S	power_d2	power_acat
dense	8	100	0.125	0.338
sparse	8	100	1.000	0.998
highdim	20	15	0.000	0.218

The **dense** case flips under correlation: with independent covariates d^2 won (0.602 vs 0.456), but with positively correlated covariates ACAT wins and d^2's power craters. This is correct: the dense tilt points in the all-covariates-equal direction, the high-variance direction of the correlated V_d, and $d^2 = \bar{d}' V_d^{-1} \bar{d}$ appropriately discounts imbalance there (it is less surprising). ACAT ignores the covariance and keeps power – the classic Hotelling-T^2 vs marginal-combination tradeoff. The **sparse** case still slightly favors d^2; the **highdim** case is decisive: d^2 has ~zero power (frozen near $p = 0.45$; the package warns "Degrees of freedom exceeds units less number of strata") while ACAT keeps power. Net: ACAT is the more robust omnibus across correlation and dimension.

```
op <- par(mfrow = c(1, 2), mar = c(5, 4, 3, 1))
bp <- rbind(`d^2` = b4$power_d2, ACAT = b4$power_acat); colnames(bp) <- b4$scenario
barplot(bp, beside = TRUE, ylim = c(0, 1), legend.text = TRUE,
        args.legend = list(x = "topright", bty = "n"),
        ylab = "power (alpha=.05)", main = "B4: d^2 vs ACAT power")
b1bar <- c(`d^2 all` = rejection_rate(b1$p_all),
           `d^2 screen` = rejection_rate(b1$p_scr), `ACAT` = rejection_rate(b1$p_acat))
barplot(b1bar, ylim = c(0, 1), col = "grey80", ylab = "power to detect x3 (alpha=.05)",
        main = "B1: detect a confound with 2\nnear-degenerate covariates")
```

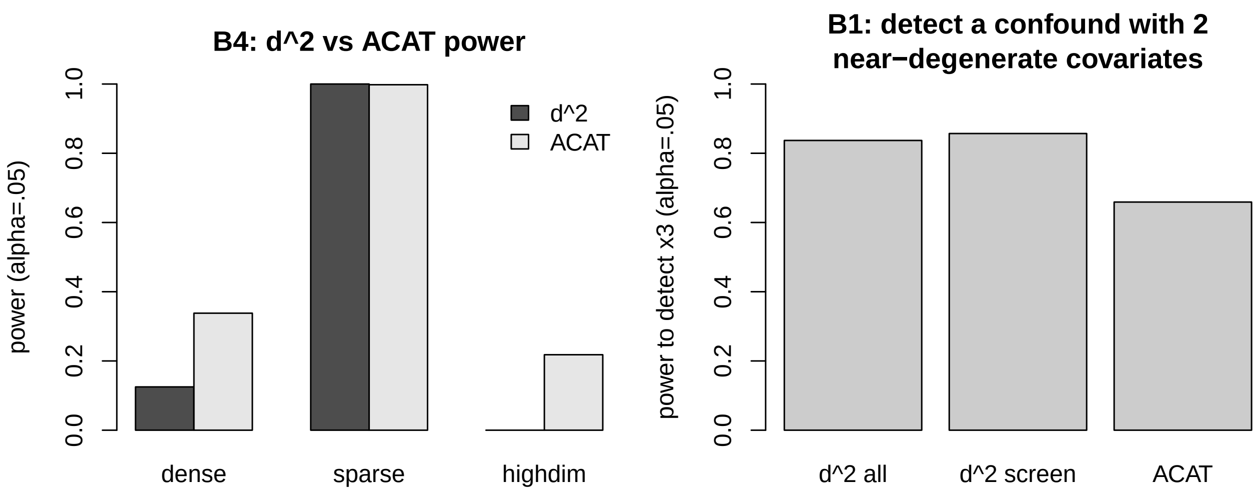

Figure 2: Study 2. Left: d^2 vs ACAT power across dense/sparse/high-dim (under correlation, ACAT wins dense and is the only one with power in high dim). Right: detecting a real confound with two near-degenerate covariates present (design screening sharpens d^2).

```
par(op)
```

B5 – the design screen cannot be gamed

```
set.seed(5)
df5 <- make_pairs(80, c(1e-4, 1, 1), c(0, 0.3, 0))
w_obs <- within_var_fraction(df5)
w_reps <- replicate(50, {
  flip <- rep(rbinom(nlevels(df5$pair), 1, 0.5) == 1, each = 2)
  within_var_fraction(within(df5, z <- ifelse(flip, 1L - z, z)))
})
```

```r
round(w_obs, 6)
#>       x1       x2       x3
#> 0.000000 0.415173 0.433022
cat(sprintf("max |w_j(re-randomized) - w_j(observed)| over 50 re-flips: %.2e\n",
            max(abs(w_reps - w_obs))))
#> max |w_j(re-randomized) - w_j(observed)| over 50 re-flips: 0.00e+00
```

w_j does not depend on the treatment vector, so an analyst cannot manipulate which covariates are screened by choosing an assignment. A screen based on the observed z-statistic would not have this property.

B8 – how small can the magnitude-screen threshold be? (the floor)

The Love-plot screen (B7) tracks balance, but its threshold cannot be made arbitrarily small. As $\tau \to 0$, the decision "balanced iff all $|\text{SMD}| < \tau$" converges to "balanced iff $\text{SMD} = 0$" – testing the sharp null of exact balance, which is exactly where the collapse lives. So there is a floor. On a BALANCED design ($\mu = 0$) the observed SMD is pure sampling noise; the screen should rarely fire. The quantity that matters is $\max|\text{SMD}|$ across the k covariates – a balanced design is "flagged" if any covariate trips the threshold (family-wise across k).

```r
maxabs_smd_balanced <- function(S, k, rho = 0.5) {
  df <- make_pairs(S, rep(1, k), rep(0, k), rho = rho)   # balanced: mu = 0
  max(abs(vapply(covariate_names(df), function(cn) smd(df, cn), numeric(1))))
}
set.seed(20260619)
b8 <- do.call(rbind, lapply(c(25, 50, 100, 200, 400), function(S) {
  m5  <- replicate(NREP, maxabs_smd_balanced(S, 5))
  m20 <- replicate(NREP, maxabs_smd_balanced(S, 20))
  data.frame(ESS = S,
             floor_k5        = round(as.numeric(quantile(m5,  0.95)), 3),
             flag_at_0.25_k5 = round(mean(m5  >= 0.25), 3),
             floor_k20        = round(as.numeric(quantile(m20, 0.95)), 3),
             flag_at_0.25_k20 = round(mean(m20 >= 0.25), 3))
}))
b8
```

ESS	floor_k5	flag_at_0.25_k5	floor_k20	flag_at_0.25_k20
25	0.462	0.491	0.565	0.859
50	0.334	0.196	0.376	0.429
100	0.230	0.024	0.258	0.064
200	0.167	0.001	0.184	0.000
400	0.116	0.000	0.131	0.000

`floor` is the threshold that yields a 5% false-flag rate (the 95th percentile of $\max|\text{SMD}|$ under balance); `flag_at_0.25` is the chance a balanced design is flagged at $\tau = 0.25$.

```r
set.seed(20260619)
taus <- seq(0, 0.6, by = 0.02)
plot(NA, xlim = c(0, 0.6), ylim = c(0, 1),
     xlab = "magnitude-screen threshold tau (SMD)",
     ylab = "P(flag a balanced design)",
     main = "B8: SMD-screen false-flag rate vs threshold (k=5)")
cols <- c("25" = "red", "100" = "black", "400" = "blue")
for (S in c(25, 100, 400)) {
  m  <- replicate(NREP, maxabs_smd_balanced(S, 5))
  ff <- vapply(taus, function(tt) mean(m >= tt), numeric(1))
  lines(taus, ff, lwd = 2, col = cols[as.character(S)])
}
abline(v = 0.25, lty = 2, col = "grey40"); abline(h = 0.05, lty = 3, col = "grey60")
legend("topright", c("ESS=25", "ESS=100", "ESS=400", "tau=0.25"),
       col = c("red", "black", "blue", "grey40"),
       lty = c(1, 1, 1, 2), lwd = c(2, 2, 2, 1), bty = "n")
```

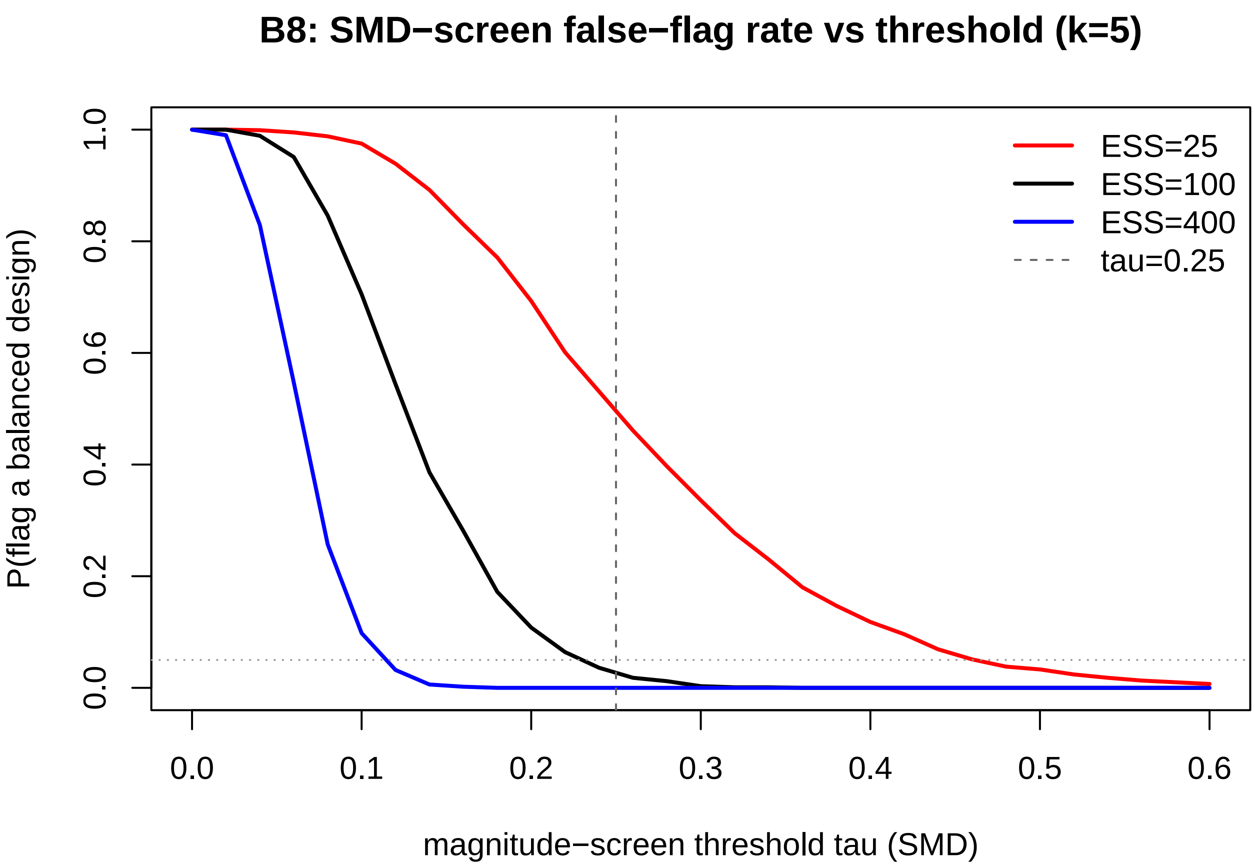

Figure 3: B8. Magnitude-screen false-flag rate on a BALANCED design vs the threshold tau (k=5). As tau -> 0 the screen flags balanced designs ever more often (its version of the collapse). The curve shifts left as ESS grows (floor ~ 1/sqrt(ESS)). The 0.25 rule of thumb (dashed) is safe at large ESS but not small.

What it means. The floor behaves as

$$\tau_{\text{floor}} \;\approx\; \rho_{\text{match}} \frac{\sqrt{2\ln k}}{\sqrt{\text{ESS}}}, \qquad \rho_{\text{match}} = \frac{\text{rms within-pair difference}}{\text{pooled SD}},$$

so it (i) shrinks like $1/\sqrt{\text{ESS}}$ – more data lets a smaller threshold still work; (ii) grows like $\sqrt{\ln k}$ with the number of covariates, through multiplicity (more covariates -> more chances for one to trip τ by noise); and (iii) shrinks with matching tightness (ρ_{match}). These are loose random pairs ($\rho_{\text{match}} \approx 0.9$, the worst case); a genuinely tight match lowers the floor and makes 0.25 safer still.

The practical reading. The threshold has TWO determinants that should not be conflated. The SUBSTANTIVE margin (how much standardized imbalance you are willing to call "balanced") is an effect-size judgment, NOT a function of sample size – 0.25 SD is the conventional margin in the matching literature (Rubin 2001 / Stuart 2010; see the provenance section below). The STATISTICAL floor above is a function of ESS, k, and matching tightness. Use the larger of the two. For typical designs (ESS in the hundreds, $k \sim 10$) the floor is ~0.05-0.15, comfortably below 0.25, so the substantive margin governs and a fixed 0.25 is fine. The floor only BINDS – and 0.25 becomes unreliable – in the small-ESS / many-covariate corner (e.g. ESS=25, $k = 20$ above), where the data cannot certify balance at the 0.25 margin. This is the equivalence-testing answer: fix the substantive margin, and whether you can CONCLUDE balance at it depends on ESS (power) and k (multiplicity) (Hartman & Hidalgo 2018). A sensible package default: flag at $|\text{SMD}| \geq 0.25$, and warn when the floor for the design's ESS and k approaches 0.25 (the screen cannot then reliably distinguish 0.25-imbalance from noise).

Provenance of the 0.25-SD rule

A literature check (2026-06-17) corrected a misattribution we had been repeating. We had called the 0.25 cutoff the "Rosenbaum-Rubin rule." That is wrong in two ways, and the threshold and denominator are not as settled as a single "0.25" suggests.

- The "0.25" in **Rosenbaum & Rubin (1985, *The American Statistician* 39:33-38)** is a CALIPER width on the linear propensity score (a matching tolerance), NOT a balance threshold. Stuart (2010) quotes them: "Rosenbaum and Rubin (1985b) generally suggest a caliper of 0.25 standard deviations of the linear propensity score." Citing R&R 1985 for a balance cutoff is a misattribution.
- The 0.25 BALANCE threshold comes from **Rubin (2001, *Health Services & Outcomes Research Methodology* 2:169-188)** as restated by **Stuart (2010, *Statistical Science* 25:1-21)**: "the absolute standardized differences of means should be less than 0.25 … (Rubin, 2001)." Rubin's own prose does not say 0.25 – his Condition 1 is "less than half a standard deviation," and for the PROPENSITY SCORE mean difference, not individual covariates. So the flat "0.25" is Stuart's compression.
- TWO CAMPS ON THE NUMBER. Matching / political-methodology tradition (Rubin 2001; Stuart 2010; Ho-Imai-King-Stuart 2007 / MatchIt) uses **0.25**. Pharmaco- epidemiology / medical tradition (Normand et al. 2001, *J Clin Epi* 54:387-398; Austin 2009, *Stat Med* 28:3083-3107; Austin 2011, *Multivar Behav Res* 46:399-424) uses **0.1** ("a standardized difference less than 0.1 … indicates a negligible difference," Austin 2011). Cohen's d "small" = 0.2 is a separate lineage. The 0.1 vs 0.25 split is a real, deliberate disagreement.

- TWO CAMPS ON THE DENOMINATOR. Stuart / MatchIt use the full TREATED-group SD fixed at pre-matching; Austin / medical use the POOLED SD $\sqrt{(s_t^2 + s_c^2)/2}$. **RItools' `std.diff` uses the pooled SD** – the Austin convention, which traditionally pairs with 0.1, not 0.25. So a 0.25 cutoff on RItools' pooled-SD `std.diff` mixes conventions; the default value AND denominator deserve an explicit, documented choice (the `covariate.scales` argument can fix the denominator to pre-matching SDs if we want the Stuart convention).
- PRIOR ART FOR OUR CENTRAL STANCE. **Ho, Imai, King & Stuart (2007, *Political Analysis* 15:199-236)** argue balance is an in-sample property and that hypothesis tests / p-values should NOT be used as balance criteria, because rejection depends on sample size (which matching changes). That is exactly the large-sample point in A3, and it supports presenting the omnibus as a direction test while leading with the magnitude (`std.diff`).

Confidence: verbatim quotes were confirmed from the primary text of Rubin (2001), Stuart (2010), Austin (2009, 2011). The R&R (1985) caliper wording and Normand et al. (2001) are from reliable secondary quotation (originals paywalled); confirm those two primaries directly before citing them in a paper.

Conclusions / package implications

1. **The collapse is real, exact, and unescapable** by any statistic on the within-set permutation reference – d^2 and ACAT alike (A1, B3), and robust to correlation. The omnibus reads the systematic tilt relative to the within-pair noise, scaled by sample size; it does not read the size of imbalance, and in large samples rejects on trivial bias (A3). For users: report a fixed-scale magnitude or equivalence result as the primary balance summary, and present the omnibus as a test of systematic direction, not a balance-quality score.

2. **Two screens, two jobs – keep them separate.**

   (a) *Design screen* (drop exactly-matched directions): build it for honest abstention plus numerical safety (B1b), a modest power gain (B1), size-preserving (B2), ungameable (B5) – but NOT a collapse fix (B3). It must work on **directions** (singular values of V_d), not raw covariates, because under correlation the degenerate direction is a linear combination w_j misses (B6). Implementation: a relative tolerance at `R/Design.R:991` on the eigenvalues of V_d relative to the complete-randomization covariance, reported as a profile over `tau`, with graceful degradation replacing the `stop()` at `R/utils.R:381`.

   (b) *Magnitude screen* (the Love-plot / 0.25-SD rule): this is the one that tracks balance and escapes the collapse (B7). It is the primary balance **report**, not a test pre-filter (assignment-dependent; would inflate the omnibus if fed to it). The package should present per-covariate SMDs against a 0.25 reference alongside, not inside, the omnibus.

3. **Keep d^2 AND switch to ACAT when needed (B4)** – stronger under correlation. ACAT wins the dense case once covariates are correlated and is the only one with power in high dimension (where d^2 freezes near 0.45). Concrete trigger: switch when #covariates (after screening) $\geq$ rank(V_d); consider preferring ACAT or reporting both more broadly given the dense-case flip. Neither omnibus escapes the collapse, so this is about power and numerical robustness, not about seeing the size of imbalance.

Open items

- Vary the correlation: sweep `rho` (and try AR(1)) to map how the B4 dense flip and the d^2/ACAT crossover depend on correlation strength and structure.
- Production direction screen: derive eigenvalues from the engine's `tcov` plus a complete-randomization reference, not the base-R within-vs-total proxy.
- Equivalence framing for the magnitude screen: choose the default margin (0.25 SD in the matching tradition vs 0.1 in the medical tradition – see provenance section) and whether to report a formal equivalence test (Hartman & Hidalgo 2018) or just SMDs vs the reference line. Also decide the denominator (RItools uses pooled SD; the 0.25 lineage uses the treated-group SD).
- If we proceed to package code: write the testthat tests FIRST (per CLAUDE_CODING.md) – these experiments are the statistical-principle tests – then implement the screen, graceful degradation, and the ACAT switch, then `devtools::check()`.

```
#> R version 4.6.0 (2026-04-24)
#> Platform: aarch64-apple-darwin23
#> Running under: macOS Tahoe 26.5.1
#>
#> Matrix products: default
#> BLAS:   /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRblas.0.dylib
#> LAPACK: /Library/Frameworks/R.framework/Versions/4.6/Resources/lib/libRlapack.dylib;  LAPACK
#>
#> locale:
#> [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
#>
#> time zone: America/Chicago
#> tzcode source: internal
#>
#> attached base packages:
#> [1] stats     graphics  grDevices utils     datasets  methods   base
#>
#> other attached packages:
#> [1] RItools_0.3-5.9001 testthat_3.3.2     survival_3.8-6     ggplot2_4.0.3
#>
#> loaded via a namespace (and not attached):
#>  [1] generics_0.1.4     tidyr_1.3.2        lattice_0.22-9     digest_0.6.39
#>  [5] magrittr_2.0.5     evaluate_1.0.5     grid_4.6.0         RColorBrewer_1.1-
3
#>  [9] pkgload_1.5.2      fastmap_1.2.0      rprojroot_2.1.1    jsonlite_2.0.0
#> [13] Matrix_1.7-5       pkgbuild_1.4.8     sessioninfo_1.2.3  brio_1.1.5
#> [17] tinytex_0.59       purrr_1.2.2        scales_1.4.0       abind_1.4-8
#> [21] cli_3.6.6          rlang_1.2.0        ellipsis_0.3.3     splines_4.6.0
#> [25] withr_3.0.2        cachem_1.1.0       yaml_2.3.12        devtools_2.5.2
#> [29] otel_0.2.0         tools_4.6.0        SparseM_1.84-2     memoise_2.0.1
#> [33] dplyr_1.2.1        vctrs_0.7.3        R6_2.6.1           lifecycle_1.0.5
#> [37] fs_2.1.0           usethis_3.2.1      pkgconfig_2.0.3    desc_1.4.3
```

```
#> [41] pillar_1.11.1      gtable_0.3.6       glue_1.8.1         xfun_0.57
#> [45] tibble_3.3.1       tidyselect_1.2.1   rstudioapi_0.18.0  svd_0.5.8
#> [49] knitr_1.51         farver_2.1.2       xtable_1.8-8       htmltools_0.5.9
#> [53] rmarkdown_2.31     compiler_4.6.0     S7_0.2.2
```